

CHOREO: Every Humanoid Skill as a Trajectory

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Abstract

Recent advances in humanoid robotics have produced diverse skills through reinforcement learning, motion imitation, and generative modeling. Yet these capabilities remain siloed because they are built around incompatible representations, interfaces, and controllers. We present CHOREO, a framework for training-free composition of heterogeneous humanoid skills. Our key observation is that, regardless of how a skill is learned, it can ultimately be expressed as an executable motion trajectory. Based on this observation, CHOREO converts each capability into SkillMotion, a unified representation that combines motion states, contacts, semantics, and boundary conditions. Skills are composed through direct continuation, cubic Hermite blending, or validated bridge motions, without retraining source models or updating models at test time. On Unitree G1 in MuJoCo, CHOREO organizes 2,950 admitted SkillMotion assets derived from heterogeneous sources and achieves 95.4% sequence success across 130 multi-action tasks, including 93.8% success on eight-action sequences. These results demonstrate that executable trajectories provide a scalable interface for accumulating and composing pretrained humanoid capabilities.

Introduction

Recent advances in reinforcement learning, motion imitation, teleoperation, and generative modeling have rapidly expanded the repertoire of humanoid robot capabilities. Modern humanoids can acquire diverse skills including locomotion, manipulation, recovery, whole-body coordination, and expressive motion generation. As increasingly powerful learning algorithms continue to emerge, the number of available humanoid skills is expected to grow substantially. However, simply accumulating independently trained capabilities does not naturally produce a more capable robot. Most existing skills remain isolated because they are developed under different learning paradigms, motion representations, controller interfaces, and execution assumptions, preventing them from being directly reused or composed into long-horizon behaviors.

A natural solution is to learn a single universal controller from increasingly diverse motion datasets. Recent work has demonstrated impressive multi-skill policies by jointly training on heterogeneous demonstrations or reinforcement learn-

ing data. While effective, these approaches fundamentally treat new capabilities as another training problem. Incorporating additional skills typically requires collecting compatible datasets, modifying the training distribution, or retraining an ever-larger policy. As humanoid capabilities are increasingly produced by different research communities, simulation platforms, and generative models, such retraining-based solutions become progressively less scalable. This raises a complementary question: **Can heterogeneous pretrained humanoid capabilities be continuously accumulated, reused, and composed without retraining their original models?**

Our key observation is that although humanoid skills originate from fundamentally different sources, they share a common behavioral consequence: every capability ultimately produces an executable robot trajectory. A reinforcement learning policy generates trajectories through interaction, motion datasets explicitly store trajectories, and generative models synthesize motions that can be converted into robot trajectories. This commonality suggests that trajectories provide a natural behavioral abstraction for organizing heterogeneous humanoid capabilities. Rather than unifying the internal architectures of different controllers, we instead unify their behavioral outputs.

Based on this insight, we present **CHOREO**, a training-free framework that organizes heterogeneous humanoid capabilities into a unified motion skill space. Instead of treating policies, motion assets, and generative models as incompatible execution modules, CHOREO converts them into reusable **SkillMotions**, enabling diverse capabilities to share a common behavioral representation while preserving their original implementations. This unified organization decouples skill generation from skill execution, allowing new capabilities to be continually incorporated without modifying existing controllers. Consequently, expanding robot behaviors becomes a process of growing a reusable motion skill library rather than repeatedly retraining increasingly complex policies.

Specifically, CHOREO addresses these challenges through three components:

- We introduce **SkillMotion**, a unified trajectory-based representation that augments robot trajectories with seman-

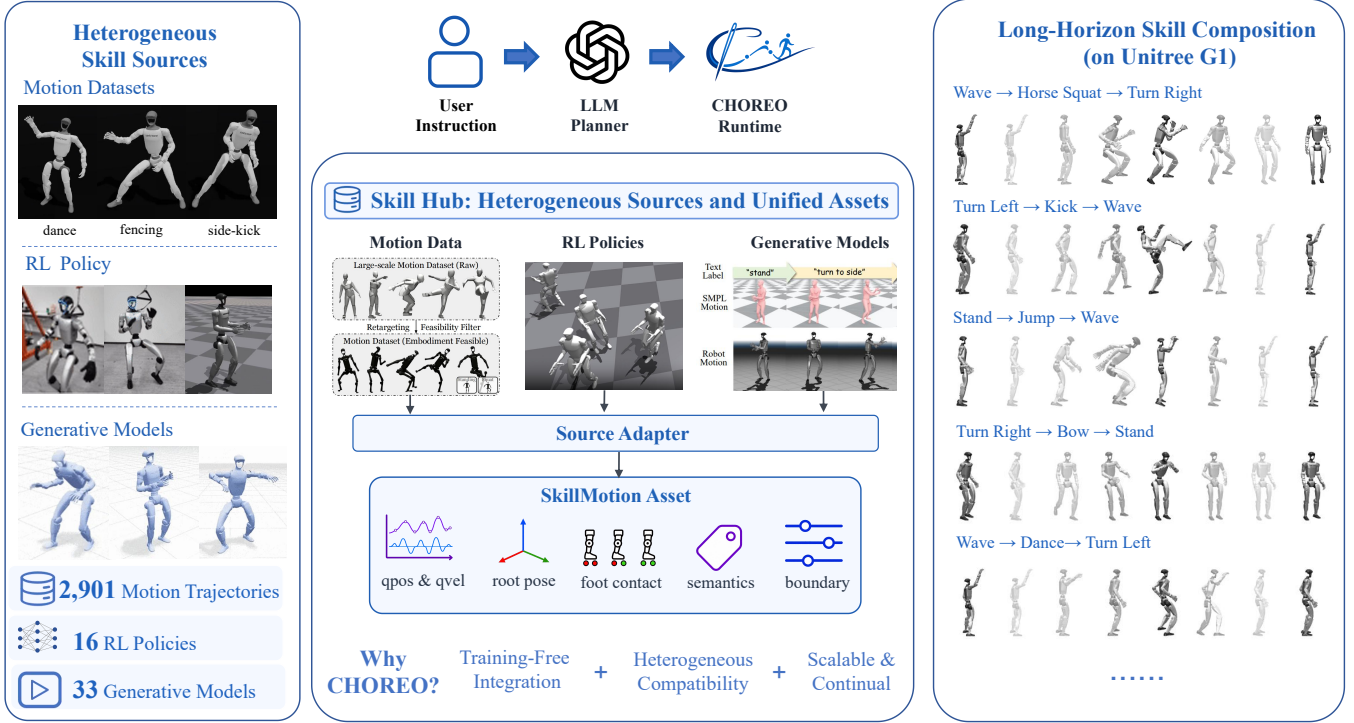


Figure 1: CHOREO converts heterogeneous skill sources—including motion datasets, RL policies, and generative models—into unified SkillMotion assets through source-specific adapters. Given a user instruction, an LLM planner selects and sequences these assets, while the CHOREO runtime executes them as long-horizon skill compositions on the Unitree G1.

tic annotations, contact states, execution constraints, and embodiment metadata.

- We develop a **tracker-aware execution mechanism** that dynamically associates each SkillMotion with a compatible tracking controller, thereby decoupling skill representation from controller implementation.
- We propose an **adaptive transition strategy** that connects compatible SkillMotions using cubic Hermite interpolation and, when the boundary gap is large, retrieves and inserts a bridge motion whose two boundaries are connected using the same interpolation procedure.

Related Work

Generalist Humanoid Control

Physics-based motion imitation allows humanoid robots to learn whole-body behaviors from motion data (Peng et al. 2018, 2021). Recent work has improved the generality of humanoid controllers along several dimensions. Large motion datasets and increased model capacity enable a single policy to track broader motion distributions (Luo et al. 2023, 2025). Flexible kinematic interfaces support different forms of motion specification (Tessler et al. 2024; He et al. 2024). Heterogeneous training data further allows one policy to cover both dynamic motions and balance-critical behaviors (Pan et al. 2025). Other works broaden the range of practical whole-body behaviors, including naturalistic locomotion, agile maneuvers, parkour, and teleoperation (Cheng et al. 2024; He

et al. 2025; Wu et al. 2026; He et al. 2024). Scaling these approaches has produced controllers that cover increasingly broad motion distributions through a common policy and execution interface. In this work, we focus on expanding the capabilities available to a humanoid system without consolidating them inside a single learned controller. CHOREO converts skills from independently developed systems into a common robot-space trajectory format called SkillMotion. This common format allows the system to retrieve, reuse, and compose skills from different sources.

Reusable Skill Interfaces and Robotic Harnesses

Robot capabilities are commonly exposed either through a shared controller or through an external runtime. Learned latent spaces support downstream control, while motion- and task-space commands allow different input sources or planners to drive a common execution policy (Peng et al. 2022; Luo et al. 2024; Tessler et al. 2024; He et al. 2024; Yang et al. 2026; Luo et al. 2026). Recent robotic harnesses instead organize tools, controllers, and policies as callable components. They use execution feedback to support capability selection, replanning, and handoffs (Liu et al. 2026; Zhang et al. 2026; Wang et al. 2026; Huang et al. 2026). These systems mainly address task-level orchestration, often in manipulation or mobile manipulation.

CHOREO brings this model-external approach to dynamic humanoid motion. Its skills may come from controllers, motion generators, demonstrations, or motion datasets, and

therefore may not share an online policy interface. CHOREO converts their motion outputs into standardized SkillMotion assets. Explicit trajectories, contact states, and motion boundaries make heterogeneous whole-body skills inspectable and composable before execution by a shared motion tracker.

Long-Horizon Humanoid Skill Composition

Long-horizon humanoid behavior requires both semantic sequencing and physically feasible transitions. Motion graphs and motion matching connect clips within a common motion database, and PHP extends this principle to long-horizon agile parkour references (Kovar, Gleicher, and Pighin 2002; Holden et al. 2020; Wu et al. 2026). Recent methods instead learn or schedule transitions within a shared control system through skill graphs, policy gating, controller switching, or jointly designed planning and execution representations (Lau et al. 2026; Xin et al. 2026; Baek, Kim, and Ha 2026; Kim et al. 2026; Tevet et al. 2025; Yu et al. 2026). These approaches generally establish compatibility within a prepared motion library or a common control stack.

CHOREO targets skills that are heterogeneous at the source. It first converts them into a common robot-space format and then evaluates transitions using explicit whole-body pose, root motion, velocity, and contact states. Compatible skills are connected directly, while an intermediate bridge is introduced when direct connection is unsuitable. This supports cross-source composition without jointly retraining the source models or a library-wide transition policy.

Method

Problem Setup and Framework Overview

CHOREO composes skills from heterogeneous sources into long-horizon reference motions for humanoid execution. Given a natural-language instruction x , a SkillMotion library $\mathcal{L}$ constructed from skill sources $\mathcal{S}$, and pretrained motion Trackers $\mathcal{T}$, CHOREO constructs an execution program

$$\mathcal{P}(x) = (\mathcal{M}_{1:K}, k_{1:K}, \Gamma_{1:K-1}), \quad (1)$$

where $\mathcal{M}_{1:K}$ denotes the resolved SkillMotions, $k_{1:K}$ records their associated Tracker indices, and $\Gamma_{1:K-1}$ denotes the transitions between adjacent skills.

As shown in Fig. 2, CHOREO separates offline registration from online composition. During registration, source-specific motion outputs are converted into canonical trajectories, annotated, and validated before admission to $\mathcal{L}$. At runtime, an LLM task planner produces an ordered skill sequence whose labels and aliases are resolved to registered assets. The planner specifies the semantic order rather than motion trajectories or control commands. CHOREO then constructs boundary transitions and executes the composed reference through the configured Tracker interfaces.

SkillMotion: a Unified Motion Representation

Each registered skill is represented as a canonical trajectory augmented with the information required for semantic resolution, boundary-aware composition, and tracking execution:

$$\mathcal{M} = (\tau, \mathbf{z}, \mathbf{b}^{\text{in}}, \mathbf{b}^{\text{out}}, \mathbf{e}). \quad (2)$$

Here, $\tau = (\mathbf{s}_{1:T}, f)$ is a trajectory of T frames sampled at frequency f . Each state $\mathbf{s}_t$ contains joint positions and velocities, the root state, and foot-contact information. All trajectories share the same embodiment, joint order, coordinate convention, and sampling format.

The descriptor $\mathbf{z}$ records semantic labels, motion attributes, and source provenance. The boundary descriptors $\mathbf{b}^{\text{in}}$ and $\mathbf{b}^{\text{out}}$ contain structured entry and exit windows together with motion-profile summaries. The execution descriptor $\mathbf{e}$ records the target embodiment, Tracker configuration, interface requirements and validation evidence. These fields support skill resolution, transition construction, and execution without revisiting the original sources.

Offline Registration of Heterogeneous Skill Sources

CHOREO converts each source item into a pending SkillMotion candidate. Existing motion clips are decoded directly. Generative models produce motion outputs in their native environments, while reinforcement learning policies are rolled out in their corresponding simulators. The Source Adapter ingests the resulting finite-horizon body-state trajectories rather than the source model parameters or policy actions.

Trajectories are retargeted when required by the source embodiment and normalized in joint order, coordinates, units, and sampling frequency. Our implementation uses a canonical 23-DoF Unitree G1 representation at 30 Hz. Each candidate is annotated with the descriptors in Eq. (2). Missing velocities are derived from the motion sequence. Contact states are retained when available, inferred from body kinematics when possible, and otherwise marked as unknown with zero confidence.

A validation gate checks format validity, motion consistency, and physical trackability with the configured Tracker. Repairable candidates may undergo deterministic correction and revalidation. Accepted candidates enter the SkillMotion registry with their validation evidence, while failed candidates are quarantined rather than exposed to runtime execution.

Tracker-Aware Skill Composition and Execution

Individually validated skills do not necessarily form a continuously executable sequence. CHOREO uses their execution descriptors to resolve Tracker configurations and their boundary descriptors to construct transitions. The composed reference is then supplied to the configured tracking controller through a shared execution interface.

Boundary compatibility For consecutive SkillMotions $\mathcal{M}_i$ and $\mathcal{M}_{i+1}$, CHOREO aligns the successor with the predecessor in planar position and heading. Boundary states are evaluated at the selected splice endpoints within the annotated exit and entry windows. For a compared state component y , let $\Delta y_i = y_{i+1}^{\text{in}} - y_i^{\text{out}}$. The boundary mismatch is

$$d_i = \sum_{\nu \in \mathcal{F}} \frac{w_\nu}{m_\nu \sigma_\nu^2} \left\| \Delta \mathbf{y}_i^{(\nu)} \right\|_2^2 + w_c E_i^c, \quad (3)$$

where $\mathcal{F} = \{q, \dot{q}, h, v\}$ indexes joint position, joint velocity, root height, and root linear velocity. $\Delta \mathbf{y}_i^{(\nu)}$ denotes the cor-

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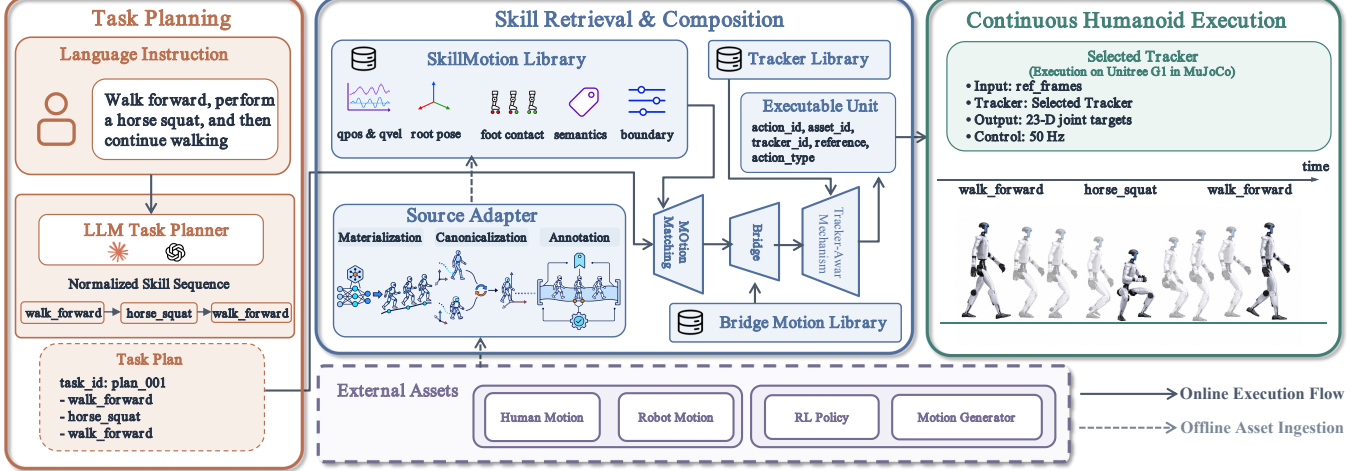


Figure 2: **Overview of CHOREO.** Offline, the Source Adapter converts heterogeneous motion sources into canonical and validated SkillMotion assets. Online, an LLM task planner produces a normalized skill sequence from a natural-language instruction, and the runtime resolves it to registered SkillMotions (**left**). CHOREO evaluates boundary compatibility, connects adjacent motions through local quintic seams, and inserts a stable-standing bridge when direct connection is unsuitable (**middle**). Tracker-specific adapters convert the composed reference into the inputs required by the tracking controller (**right**). Our experiments use GMT on the Unitree G1 at 50 Hz. Solid and dashed arrows denote online execution and offline registration, respectively.

responding boundary-state difference after alignment, and m_v is its dimension, with $m_q = m_{\dot{q}} = n$, $m_h = 1$, and $m_v = 3$. The weights w_v and scales σ_v control the contribution and normalization of each feature, while E_i^c measures confidence-weighted foot-contact mismatch, where n is the number of actuated joints, h is root height, and $\mathbf{v}$ is root linear velocity expressed in the aligned frame. The non-negative weights w control the contribution of each term, and the positive scales σ normalize quantities with different units. The contact term E_i^c is a confidence-weighted mean of foot-contact differences; unavailable contact information is omitted rather than treated as a known non-contact state. Joint-position differences measure pose mismatch, while joint and root velocities distinguish similar poses with different motion trends. Height and contact terms additionally describe vertical posture and support configuration. The score is a reference-level screening criterion, not a guarantee of physical feasibility. A boundary is eligible for direct connection when $d_i < \delta$, where δ , the normalization scales, and the weights are fixed configuration parameters.

Local transition construction For a directly connected pair, CHOREO replaces a short boundary window with a quintic seam. For endpoints belonging to motions a and b , the joint reference satisfies:

$$\mathbf{q}(t) = \sum_{r=0}^5 \alpha_r t^r, \quad t \in [0, \Delta t]. \quad (4)$$

$$\mathbf{q}^{(\ell)}(0) = \mathbf{q}_{a,\text{out}}^{(\ell)}, \quad \ell = 0, 1, 2. \quad (5)$$

$$\mathbf{q}^{(\ell)}(\Delta t) = \mathbf{q}_{b,\text{in}}^{(\ell)}, \quad \ell = 0, 1, 2. \quad (6)$$

where Δt is the seam duration and superscript (ℓ) denotes the ℓ -th time derivative. The six endpoint constraints determine the polynomial coefficients, matching joint position, velocity, and acceleration while preserving motion outside the transition window.

When direct connection is unsuitable, CHOREO inserts a pre-validated stable-standing reference τ_{st} . Unlike a local seam, which smooths a boundary, the bridge changes the intermediate reference by introducing a standing configuration. The same seam operator connects the source to the bridge and the bridge to the successor:

$$\Gamma_i = \begin{cases} \mathcal{J}_{i,i+1}, & d_i < \delta, \\ \mathcal{J}_{i,\text{st}} \oplus \tau_{\text{st}} \oplus \mathcal{J}_{\text{st},i+1}, & d_i \geq \delta, \end{cases} \quad (7)$$

where $\mathcal{J}_{a,b}$ denotes the local seam and $\oplus$ denotes temporal stitching with replaced boundary windows removed. The retained motion interiors and transition segments form the composed reference $\bar{\tau}$. Validation of the standing asset alone does not establish the trackability of the newly constructed transitions; sequence-level execution is evaluated in our experiments.

Tracker-specific execution The runtime resolves each asset against its recorded admission status and Tracker configuration. For a configured Tracker $\mathcal{T}_k$, reference and observation adapters construct the controller inputs:

$$\begin{aligned} \mathbf{r}_t &= \phi_k^{\text{ref}}(\bar{\tau}, t), \\ \mathbf{a}_t &= \pi_k(\phi_k^{\text{obs}}(\mathbf{x}_t), \mathbf{r}_t), \end{aligned} \quad (8)$$

where $\mathbf{x}_t$ is the robot state, $\mathbf{r}_t$ is the tracking reference, and $\mathbf{a}_t$ is the controller output. The reference adapter handles

coordinate conversion, temporal resampling, and reference-window construction, while the observation adapter constructs the Tracker-specific observation. Original skills, local seams, and bridge motions use the same interface.

Our experiments instantiate this interface with GMT on the Unitree G1, keeping the Tracker fixed throughout each sequence. Canonical references are stored at 30 Hz and supplied to GMT at its 50 Hz control rate. The frozen evaluation protocol uses pre-registered SkillMotions without invoking source generators or source policies during online composition or execution.

Experiments

We evaluate CHOREO along three dimensions: long-horizon skill composition, integration of heterogeneous skill sources, and compatibility with different low-level trackers. We further ablate the transition design to identify the components that support reliable long-horizon execution.

Experimental Setup and Protocol

All experiments are conducted on Unitree G1 in MuJoCo using frozen low-level policies. CHOREO uses GMT for the long-horizon and source-adaptation benchmarks, whereas the tracker benchmark evaluates each frozen tracker without parameter updates. Within each benchmark, all compared methods receive the same prompts or reference motions, initial states, motion assets, execution horizons, and failure criteria. Prompt plans are fixed before execution, while tracker-aware routing, executable recovery, online replanning, and test-time gradient updates are disabled. The long-horizon comparison and ablations use seed 20260726. Each task or clip is executed once, and every failure remains in the denominator.

The long-horizon benchmark contains 130 prompt-conditioned sequences, including 20 two-action, 26 three-action, 36 five-action, and 48 eight-action tasks, with 552 planned skill boundaries in total. The default transition window spans 24 control frames. Each episode covers the complete constituent motions and inserted transitions without an additional wall-clock cutoff.

The source-adaptation benchmark contains 2,969 motion-library trajectories, 50 RL-policy trajectories, and 50 diffusion-generated motions. All 3,069 instances undergo registration checks for schema validity, finite states, and save-load consistency. A deterministic subset of 256 library motions, together with all RL and diffusion motions, is additionally evaluated through closed-loop rollout. The tracker benchmark uses the same 30 admission-approved G1 trajectories for each tracker. Each trajectory is resampled to 50 Hz and evaluated for at most 3 s.

A sequence succeeds only if all requested skills and transitions are completed. A planned boundary counts as a successful switch when it is reached, the transition completes, and execution enters the intended target skill. Boundaries not attempted because of an earlier failure are counted as switch failures. Transition latency is reported but is not used as a success threshold. A persistent fall is recorded when the failure condition holds for 25 consecutive control frames,

corresponding to 0.5 s at a 0.02 s control step. We report Δq as the mean, over attempted boundaries, of the maximum adjacent joint-position jump at each executed seam.

For source adaptation, import success requires conversion into a finite and valid SkillMotion asset, whereas execution success additionally requires admission and completion of the full physical rollout. Tracker success requires completing the evaluated reference trajectory with finite states and without a persistent fall.

Long-Horizon Skill Composition

We evaluate whether CHOREO maintains reliable execution as sequence length and the number of skill boundaries increase. We compare it with direct switching, fixed Hermite interpolation, a fixed standing bridge, and motion matching on the same 130 sequences. All methods use the same task sequences, motion assets, initial states, and frozen GMT tracker. The comparison therefore isolates the effect of transition construction while holding the constituent skills and low-level controller fixed. As shown in Table 1, CHOREO completes 124 of the 130 sequences, corresponding to an overall success rate of 95.4%. It completes all 46 two- and three-action sequences and remains above 90% on the longer subsets, completing 33/36 five-action and 45/48 eight-action sequences. The strongest baseline reaches 63.9% on five-action sequences and 41.7% on eight-action sequences. Consequently, CHOREO’s advantage over the best-performing baseline increases from 15.0 percentage points on two-action tasks to 52.1 points on eight-action tasks.

This widening advantage is especially important under the benchmark’s sequence-level success criterion. A sequence is considered successful only when every requested skill and every intermediate transition is completed. Longer sequences therefore expose each method to repeated opportunities for a boundary error to terminate the entire execution. CHOREO’s 93.8% success on eight-action sequences shows that its transition reliability is largely preserved even after multiple skill boundaries, whereas the strongest baseline succeeds on fewer than half of these sequences.

The transition-level metrics provide a consistent explanation for this difference. Across the 552 planned skill boundaries, CHOREO achieves a switch success rate of 96.7%, exceeding Motion Matching, the strongest alternative on this metric, by 21.5 percentage points. It also reduces the fall rate from 41.5% for the safest baseline to 4.6%, a reduction of 36.9 percentage points. At the same time, CHOREO records the lowest joint-position discontinuity at 0.0170 rad, essentially matching Motion Matching at 0.0171 rad.

The baseline results further show that smooth seams alone are insufficient for reliable composition. Fixed Hermite interpolation reduces Δq from 0.0449 rad under direct switching to 0.0178 rad, yet its eight-action success remains at 22.9% and its fall rate remains above 50%. A fixed standing bridge performs better on long sequences, reaching 41.7% success on eight-action tasks, but its 41.5% fall rate shows that a single intermediate posture is insufficient for the range of boundaries in this benchmark. Motion Matching achieves similarly low discontinuity, but its switch and sequence success remain substantially below those of CHOREO.

Table 1: Long-horizon skill-composition results on 130 fixed sequences. Sequence success requires completion of every constituent skill and transition. Higher success rates are better; lower fall rates and joint-position discontinuities are better.

Method	2-Action SR	3-Action SR	5-Action SR	8-Action SR	Switch SR	Fall Rate	Δq (rad)
Direct Switching	75.0%	61.5%	58.3%	14.6%	67.2%	54.6%	0.0449
Fixed Hermite	85.0%	69.2%	44.4%	22.9%	68.8%	52.3%	0.0178
Fixed Stand Bridge	70.0%	73.1%	63.9%	41.7%	72.6%	41.5%	0.0180
Motion Matching	80.0%	73.1%	61.1%	31.2%	75.2%	44.6%	0.0171
CHOREO	100.0%	100.0%	91.7%	93.8%	96.7%	4.6%	0.0170

Together, these results show that CHOREO does not improve long-horizon execution merely by smoothing individual seams. Its main advantage is the ability to maintain boundary feasibility, low fall rates, and motion continuity simultaneously, preventing transition errors from accumulating across repeated skill compositions.

Heterogeneous Skill Sources

We next test whether the common skill interface can absorb outputs with substantially different structures: joint-space trajectories from a motion library, rollouts from a pretrained RL policy, and motions generated by a diffusion model. We report import and execution separately because successful conversion alone does not guarantee that an asset remains physically executable after normalization.

Table 2: Source adaptation across heterogeneous skill sources. Execution requires successful admission and rollout with frozen GMT.

Source	Import	Execution	Exec. SR
Motion Library	2969 / 2969	253 / 256	98.8%
RL Policy	50 / 50	50 / 50	100.0%
Diffusion Motion	50 / 50	50 / 50	100.0%
Total	3069 / 3069	353 / 356	99.2%

All 3,069 instances are converted into valid SkillMotion assets. Among the 356 cases evaluated through physical rollout, 353 execute successfully, yielding a success rate of 99.2%. The three failures occur in the sampled motion-library subset, while all evaluated RL-policy and diffusion-generated motions complete successfully. The experiment therefore separates two distinct requirements: import verifies that heterogeneous source outputs can be converted into the common representation, whereas rollout evaluates whether the resulting assets remain executable through the same frozen controller.

The 100% rollout success of the RL-policy and diffusion subsets shows that the interface is not tied to curated motion-library trajectories or to a single upstream representation. These sources are incorporated without retraining either their source models or GMT. The reported 99.2% execution rate is computed over the 356 rollout-tested instances rather than all 3,069 imported assets.

Multi-Tracker Compatibility

We evaluate whether the same registered SkillMotion representation can be executed by trackers with different native observation and reference interfaces. Each frozen tracker receives the same 30 admission-approved G1 trajectories through its tracker-specific adapter. Language planning, motion generation, recovery, tracker routing, and parameter updates are disabled. This experiment therefore evaluates cross-tracker execution compatibility rather than the end-to-end capabilities of the systems from which these trackers originate. Through CHOREO’s tracker-specific adapters, GMT, Holo-

Table 3: Compatibility of the SkillMotion interface with six frozen trackers on 30 shared G1 trajectories.

Frozen Tracker	Success	Success Rate
GMT	30/30	100.0%
OpenTrack	14/30	46.7%
TWIST2	26/30	86.7%
HoloMotion	30/30	100.0%
SONIC	30/30	100.0%
H-ACT	30/30	100.0%

Motion, SONIC, and H-ACT each complete all 30 trajectories. TWIST2 completes 26/30 trajectories, whereas OpenTrack completes 14/30. These results show that the registered SkillMotion assets can be consumed by trackers with substantially different native interfaces without modifying the motion assets themselves. Because four trackers reach 100.0% success, this benchmark should be interpreted as evidence of interface compatibility rather than as a ranking of complete systems. The performance range from 46.7% to 100.0% also shows that a common motion representation does not remove differences in low-level tracking robustness.

Transition Component Ablation

Finally, we isolate the contributions of motion-continuity scoring, controller-compatibility scoring, and bridge validation. All four variants reuse the same 130-task manifest, prompt plans, motion assets, initial states, seed, frozen GMT tracker, and evaluation criteria. The experiment is executed as a separately validated CUDA-sharded run, so we restrict comparisons to variants within this ablation. Across the four configurations, all 520 episodes pass independent record-integrity validation with no duplicate trials or runtime/resource failures; all observed execution failures are physical falls.

Independent validation confirms 520 unique episode records with no runtime or resource failures; all observed execution failures are physical falls.

The w/o Continuity variant removes weighted boundary and bridge costs from candidate ranking while retaining hard feasibility gates. The w/o Compatibility variant replaces state-conditioned entry matching with the default static entry while keeping hard compatibility constraints active. The w/o Bridge Validation variant preserves bridge construction and scoring but bypasses rejection and admission checks. All other components remain unchanged. We report sequence success on the longer five- and eight-action subsets and aggregate transition metrics over the complete benchmark.

Table 4 identifies state-entry compatibility as the most in-

Table 4: Transition-component ablation on 130 fixed tasks. Higher success rates are better; lower fall rates and Δq are better.

Variant	5-Act. SR	8-Act. SR	Fall Rate	Δq (rad)
Full CHOREO	91.7%	93.8%	5.4%	0.0162
w/o				
Continuity	88.9%	95.8%	6.2%	0.0177
w/o				
Compatibility	86.1%	75.0%	16.2%	0.0166
w/o				
Bridge Validation	88.9%	95.8%	5.4%	0.0161

fluent transition component. Removing it reduces eight-action success from 95.8% to 75.0% and switch success from 97.1% to 90.6%, while increasing the fall rate from 5.4% to 16.2%. These results indicate that matching the current robot state to an appropriate skill entry becomes increasingly important as transition errors accumulate over long sequences. Removing continuity scoring leaves five- and eight-action success unchanged, but increases Δq from 0.0162 to 0.0177 rad and the fall rate from 5.4% to 6.2%. Its measurable contribution is therefore primarily smoother and safer transitions. Disabling bridge validation does not reduce sequence success or increase the fall rate on this fixed benchmark. This result should not be interpreted as showing that validation is unnecessary, because its safety role concerns rejected bridge candidates that may be underrepresented in the evaluated task set.

Conclusion

We presented CHOREO, a runtime framework for composing heterogeneous humanoid motion resources and independently trained controllers into long-horizon behaviors. CHOREO converts skills from motion libraries, RL policies, and generative models into a common executable representation, and resolves transitions by jointly considering motion continuity and controller compatibility. On the 130-task benchmark, CHOREO maintains success rates above 90% for both five- and eight-action sequences while substantially reducing transition failures and falls. It also converts all 3,069 evaluated source instances, successfully executes 353 of 356 rollout-tested motions, and supports execution through mul-

tiply frozen trackers without retraining their source models or low-level policies. These results support modular skill reuse and runtime composition as a practical approach to scaling humanoid capabilities beyond monolithic policy training.

Limitations and Future Work

The conclusions of this study should be interpreted within the scope of the current evaluation. The long-horizon benchmark contains fixed tasks of up to eight actions, with each task executed once; consequently, performance variance, substantially longer horizons, and generalization to unseen task distributions remain insufficiently characterized. Although all 3,069 source instances pass registration, only 356 are further evaluated through closed-loop rollout. The tracker comparison is also limited to 30 admission-approved trajectories and measures motion-execution compatibility rather than the complete planning and language capabilities of the associated systems. Moreover, all experiments are conducted in simulation and therefore do not fully capture sensing noise, actuation delay, model mismatch, contact uncertainty, or external disturbances on physical robots.

CHOREO is further bounded by the coverage and quality of its available skill and bridge libraries, while its current transition score relies on manually designed kinematic and contact features. Future work will expand execution validation to larger and more diverse skill sets, evaluate repeated trials and longer out-of-distribution compositions, and deploy the framework on physical humanoids. We will also investigate learned transition-feasibility models, state-conditioned bridge generation, automated skill onboarding and tracker validation, and closed-loop recovery and adaptation under real-world uncertainty.

Acknowledgments

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